# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

# BIOLOGY

Past Paper 2024 • Paper Code: 4461 / 4466

Retyped faithfully from the original board paper — same questions, no changes.

## GROUP – I

Q1. Q. 1

Q1. Bilirubin is a metabolic waste formed as a result of _____ breakdown.

(A) (B)

(C) (D)

Q2. Tetanus is caused by _____

(A) (B)

(C) (D)

Q3. Slighlly elastic connective tissue that holds the Bones together is called.

(A) (B)

(C) (D)

Q4. The form of learning that involves diminution in response as a result of repeated stimuli.

(A) (B)

(C) (D)

Q5. Plants become etiolated when grown without

(A) (B)

(C) (D)

Q6. Which one of these hormone is not related to ovarian cycle?

(A) (B)

(C) (D)

Q7. During ascidian development, gut is formed by _____ cytoplasm.

(A) (B)

(C) (D)

Q8. Chromosomes were discovered by _____ in 1882.

(A) (B)

(C) (D)

Q9. In Okazaki fragments range from 1000-2000 nucleotides

(A) (B)

(C) (D)

Q10. In human cell cycle, _____ takes the least time to complete.

(A) (B)

(C) (D)

Q11. Red color blindness is also known as

(A) (B)

(C) (D)

Q12. Recombinant DNA is introduced into the host cell by means of a

(A) (B)

(C) (D)

Q13. Antithrombin–III is biotechnological product produced in

(A) (B)

(C) (D)

Q14. Which one of these is the ultimate source of all changes.

(A) (B)

(C) (D)

Q15. Third stage of xerosere is known as

(A) (B)

(C) (D)

Q16. Macaca mulatta is biological name of

(A) (B)

(C) (D)

Q17. Which one of these is non-renewable source of energy on earth?

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) On a cool day a human's temperature may be several degrees lower in arms and legs as compared to trunk, why?

(ii) Write structural formula of Urea and Uric Acid.

(iii) Is liver a major homeostatic organ? Justify in few lines.

(iv) Why calcium ions are basic requirment for muscle contraction?

(v) How is turgor pressure built in a plant cell?

(vi) What do you understand by antagonistic arrangement of muscles? Give example.

(vii) How is reproduction significant for the survival of a species?

(viii) Suggest a remedy for the parents which are unable to enjoy normal process of fertilization and birth.

(ix) What do you understand by the productivity of an aquatic ecosystem?

(x) What was the reason of desertification in Sahel at southren edge of Sahara desert?

(xi) How can we minimize the effects of energy shortage?

(xii) What do you understand by the term "Global warming"?

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) Describe the structure of spinal cord.

(ii) Differentiate between somatic and autonomic nervous system.

(iii) Give some differences between etiolation chlorosis.

(iv) What happens when a human is given wrong blood transfusion?

(v) What pattern of sex-determination is found in grasshopper? Elaborate.

(vi) Describe sexual dimorphism in drosophila.

(vii) Is it possible to extract metals from low graded ores using Bioteconology? How?

(viii) How a suspected rapist can be identified?

(ix) What are molecular scissors? Give examples.

(x) What is commensalism? Give example.

(xi) What is denitrification? Write its impact.

(xii) Differentiate between habitat and niche.

Q4. (Marks 6 × 2 = 12) 4. Answer briefly any Six parts from the followings:-

(i) How is a blastula formed in a developing chick embryo?

(ii) Define Teratology. Enlist any two causes of abnormal development.

(iii) What will happen to replication of DNA, if primase is not present.

(iv) Where codon and anticodon are situated.

(v) How is translation terminated?

(vi) How do Karyokinesis and cytokinesis phases of cell division differ?

(vii) Enlist four important functions of Mitosis.

(viii) What is genetic drift? How does it affects gene frequency?

(ix) How artificial selection is different from natural selection.

Q. Section ---------- II

Note: Attempt any three questions. (8 × 3 = 24)

Q5. 5. (a) What are Nephrons? Explain with the help of labelled diagram?

(b) Explain various stages of Prophase I.

Q6. 6. (a) Define tropic movements. Explain its different types.

(b) How energy flows in Food Chain of an ecosystem.

Q7. 7. (a) What is feed back mechanism? Explain with the help of an example.

(b) Define and explain Hardy Weinberg theorem.

Q8. 8. (a) Explain in detail the process of birth in human female.

(b) What are sex-chromosomes? Discuss the chromosomal patterns of sex determination in animals.

Q9. 9. (a) Define meristems? Discuss their various types?

(b) What is gene therapy, Give its types and Explain in which disease ex-vivo-gene therapy is needed?

# GROUP – II

Q1. Q. 1

Q1. In which zone of Lake ecosystem, light does not peneterate? up to the bottom.

(A) (B)

(C) (D)

Q2. The natural heat energy trapped under ground is called

(A) (B)

(C) (D)

Q3. An animal living in fresh water is more likely to excrete its waste nitrogen in the form of

(A) (B)

(C) (D)

Q4. Locomotory structures found in star fish are

(A) (B)

(C) (D)

Q5. Presence of irregular stripes is the character of which type of Muscles?

(A) (B)

(C) (D)

Q6. Which growth hormone can be sprayed on the tree crops to regulate fruit drop at the end of the season?

(A) (B)

(C) (D)

Q7. An individual has exopthalamic goiter and abnormally high basal metabolic rate, is more likely to be suffering from

(A) (B)

(C) (D)

Q8. Which harmones are secreted by Mammalian Placenta?

(A) (B)

(C) (D)

Q9. In which phase of Animal development migration and rearrangement of cells occur to form three germ layers.

(A) (B)

(C) (D)

Q10. What is the role of enzyme DNA ligase during DNA replication?

(A) (B)

(C) (D)

Q11. Enzyme Amino acyl tRNA synthetase has an important role during Translation.

(A) (B)

(C) (D)

Q12. Which phase of mitosis ensures equal distribution of chromosomes in the daughter cells?

(A) (B)

(C) (D)

Q13. A colour blind man is married to normal female, what is the risk of colour blind child in this family?

(A) (B)

(C) (D)

Q14. During DNA finger printing, unique collection of various sized DNA fragments, can be obtained by

(A) (B)

(C) (D)

Q15. In Recombinant DNA technology, Bacterial cells can be made more permeable for recombinant plasmids after treating with

(A) (B)

(C) (D)

Q16. One of the following is not related to Darwinism.

(A) (B)

(C) (D)

Q17. The relationship between insects and flowering plants, is an example of

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) How plants of cold regions repond to freezing temperatures?

(ii) What is shivering thermogenesis?

(iii) Differentiate vasodilation from vasoconstriction.

(iv) What are unguligrade animals?

(v) How locomotion occurs in snail?

(vi) Define muscle fatigue.

(vii) Give importance of seed dormancy.

(viii) What are viviparous mammals? Give one example.

(ix) What are acid rains? Give two disadvantages of acid rains.

(x) What is algal bloom?

(xi) Write down the importance of grassland ecosystem.

(xii) Where Tundra ecosystem exists in Pakistan?

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) Differentiate between etiolation and chlorosis?

(ii) Write the distribution of pain and cold receptors on animal body?

(iii) Give the two commercial uses of Gibberellins?

(iv) What is test cross? Also write its significance?

(v) Define pleiotropy? Give its two examples?

(vi) Differentiate between gene linkage and linkage group?

(vii) Write two uses of PCR amplification and Ananlysis?

(viii) What are restriction endonucleases? Give their function?

(ix) Give the biotechnological uses of bacteria in mining?

(x) Define climax community with example?

(xi) Differentiate between ectoparasite and endoparasite.

(xii) Discuss the role of decomposers in ecosystem?

Q4. (Marks 6 × 2 = 12) 4. Answer briefly any Six parts from the followings:-

(i) Define growth and embryonic development.

(ii) Write the role of clear and yellow cytoplasm in development.

(iii) What are okazaki fragments? In which strand they are formed?

(iv) Name the single ring nitrogen bases, also draw it.

(v) Name transforming principle, also define term transformation.

(vi) Name Trisomic Sexual non-disjunction in your Text book, Give two symptoms.

(vii) Why and how chromosome number is halved by Meiosis.

(viii) What are Analogous organs, Give one example?

(ix) Write two preventive measures to save endangered species.

Q. Section ---------- II

Note: Attempt any three questions. (8 × 3 = 24)

Q5. 5.(a) Describe the structure of nephron with labeled diagram.

(b) Write a detailed note on cancer.

Q6. 6.(a) How does healing process proceed when a bone is broken in an accident?

(b) Define succession. Explain all stages of Xerosere.

Q7. 7.(a) Explain factors which affect gene frequency.

(b) Describe the endocrine and neural functions of hypothalamus. (two each)

Q8. 8.(a) What structure are associated with human female reproductive system. Also write their function.

(b) State Law of Segregation prove it with one suitable example.

Q9. 9.(a) How would you describe the process of growth correlations in plants?

(b) What is the methodology used for expression of Recombinant DNA?